HAT-P-36b

R-0784

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_250507` · created 2026-07-28T06:37:28+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460802.8997 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1502 +/- 0.0096
Transit depth (Rp/Rs)^2	2.26 +/- 0.29 [%]
Orbital Inclination (inc)	82.91 +/- 2.36
Airmass coefficient 1 (a1)	1.436 +/- 0.017
Airmass coefficient 2 (a2)	-0.1523 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	None
Optimal Aperture	3.62
Optimal Annulus	10.15
Transit Duration (day)	0.086 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1502 ± 0.0096 vs 0.126 (deviation 2.52σ)
Duration fit vs published	0.086 d vs 0.0925 d (deviation 0.66σ)
Mid-time O–C	10.4 min
Depth SNR	15.6
Residual scatter	1.19 %

Light curve

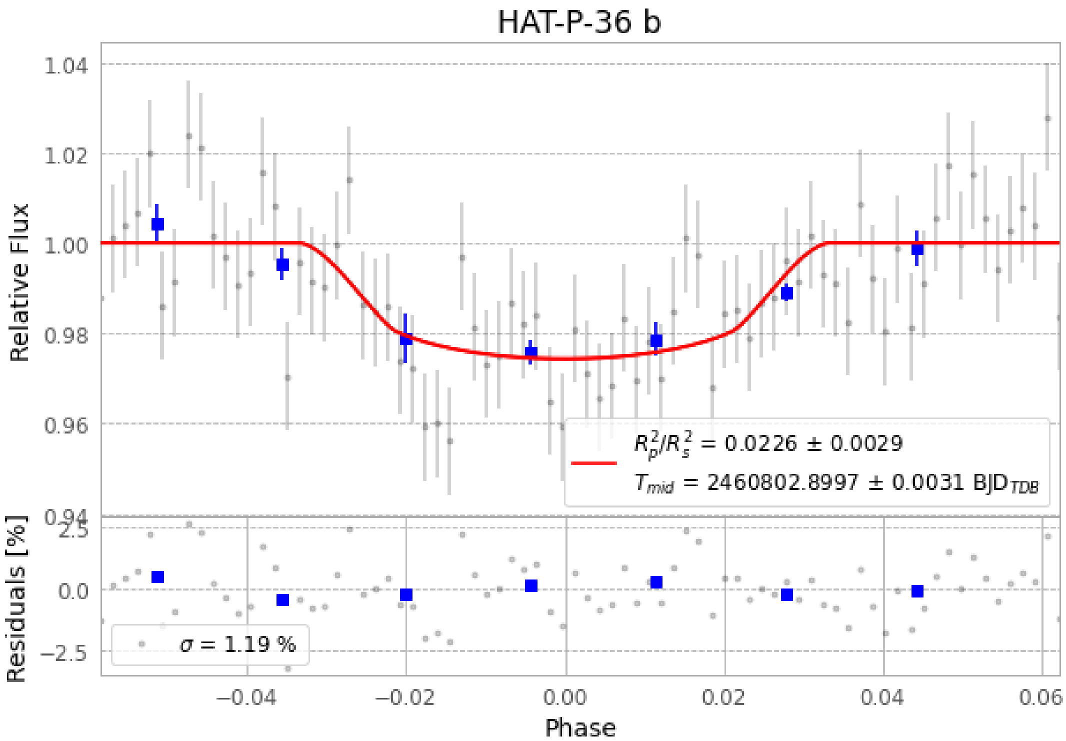

Figure 1 — FinalLightCurve_HAT-P-36 b_2025-05-07.png (SHA-256 c77d2d53fdb33131…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 289], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ad07a25c4487796d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

78 FITS frames (51 MB), `HATP-36250507073915.FITS` → `HATP-36250507113015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250507.log (252cf4e19fac…), FinalLightCurve_HAT-P-36 b_2025-05-07.png (c77d2d53fdb3…), FinalLightCurve_HAT-P-36 b_2025-05-07.csv (e0b141e64b9b…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 06:37Z.